Data Visualisation & Story Telling

Specialist Diploma in Data Analytics/Generative AI
Oct 2025 Semester

ASSIGNMENT 1
(Individual Assignment)

Submission Deadline:
13 December 2025 (Saturday), 2359 hrs

Tutorial Group :
Student Name : **Ng Chee Wee**
Student Number : **3303451Y**
Video submission link :

Table of contents

Contents

1. Project Objectives

The goal of this assignment is to design and implement an interactive dashboard using Power BI to explore the Housing and Development Board (HDB) resale flat prices dataset in Singapore, spanning 1990 to 2022.

The primary objectives are to prepare and clean the dataset for analysis, formulate meaningful exploratory questions tailored to a target audience of regular Singapore residents—who often rely on HDB flats as primary housing and investment assets—and derive actionable insights through univariate/multivariate analyses and visualizations, cumulating in interactive dashboards to answer real-world business questions. The visualisations and dashboards created in PowerBI will attempt to answer the following questions that a regular Singapore resident would care about.

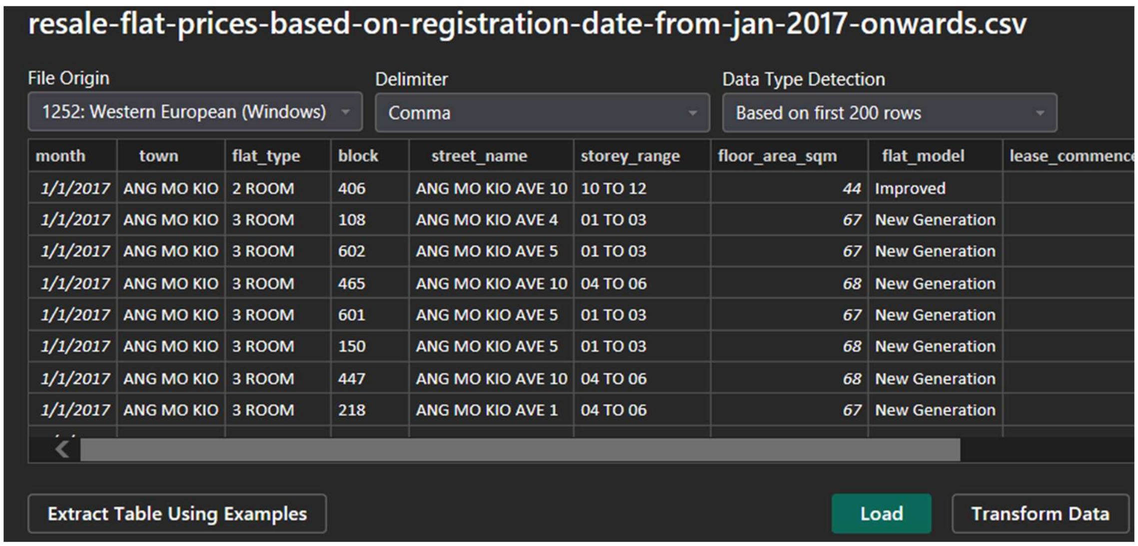
1. Explore Location-Based Demand: Identify which towns in Singapore attracted the highest number of resale transactions between 1990 and 2022, highlighting popular areas for first-time buyers.
2. Analyze Average Pricing by Location: Determine the average resale price per square feet across various towns, revealing affordability disparities and potential hotspots for value appreciation.
3. Assess Flat Type Popularity: Examine which flat types (e.g., 2-room vs. 5-room) generated the most sales volume over the dataset period,
4. Flat features: Do higher floors command a premium in resale price?
5. How has the average flat size (PSF) changed over the years?
6. How has HDB resale prices changed over the last 30 years?
7. Is there a seasonal trend in resale volume?
8. Investigate Price-lease remaining Correlation: Analyze the relationship between remaining lease and transaction price, assessing if older flats depreciate or hold value as assets.
9. Does bigger flats always means higher resale price? Investigate the correlation between floor area (sqft) and resale price

2. Data Preparation

The data set contains HDB resale flat information in Singapore from 1990 to 2022 and includes information such as when the transaction was made, town, flat type, block number, and the resale price etc. The data comes in the form of 5 separate CSV files and I will explore and investigate them individually next.

According to the metadata for resale flat prices, “Prior to March 2012, data is based on date of approval for the resale transactions. For March 2012 onwards, the data is based on date of registration for the resale transactions.”

2.1 Initial exploration and pre-processing



month	town	flat_type	block	street_name	storey_range	floor_area_sqm	flat_model	lease_commence
1/1/2017	ANG MO KIO	2 ROOM	406	ANG MO KIO AVE 10	10 TO 12	44	Improved	
1/1/2017	ANG MO KIO	3 ROOM	108	ANG MO KIO AVE 4	01 TO 03	67	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	602	ANG MO KIO AVE 5	01 TO 03	67	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	465	ANG MO KIO AVE 10	04 TO 06	68	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	601	ANG MO KIO AVE 5	01 TO 03	67	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	150	ANG MO KIO AVE 5	01 TO 03	68	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	447	ANG MO KIO AVE 10	04 TO 06	68	New Generation	
1/1/2017	ANG MO KIO	3 ROOM	218	ANG MO KIO AVE 1	04 TO 06	67	New Generation	

Import and load the data of the five csv file into PowerBI to do initial inspection, we will transform the data in the next step after inspection.

Table: resale-flat-prices-based-on-approval-date-1990-1999 (287,196 rows)

The resale flat prices 1990-1999 table contains 287,196 rows and 10 columns.

Table: resale-flat-prices-based-on-approval-date-2000-feb-2012 (369,651 rows)

The resale flat prices 2000 to Feb 2012 table contains 369,651 rows and 10 columns.

Table: resale-flat-prices-based-on-registration-date-from-mar-2012-to-dec-2014 (52,203 rows)

The resale flat prices Mar 2012 to Dec 2014 table contains 52,203 rows and 10 columns.

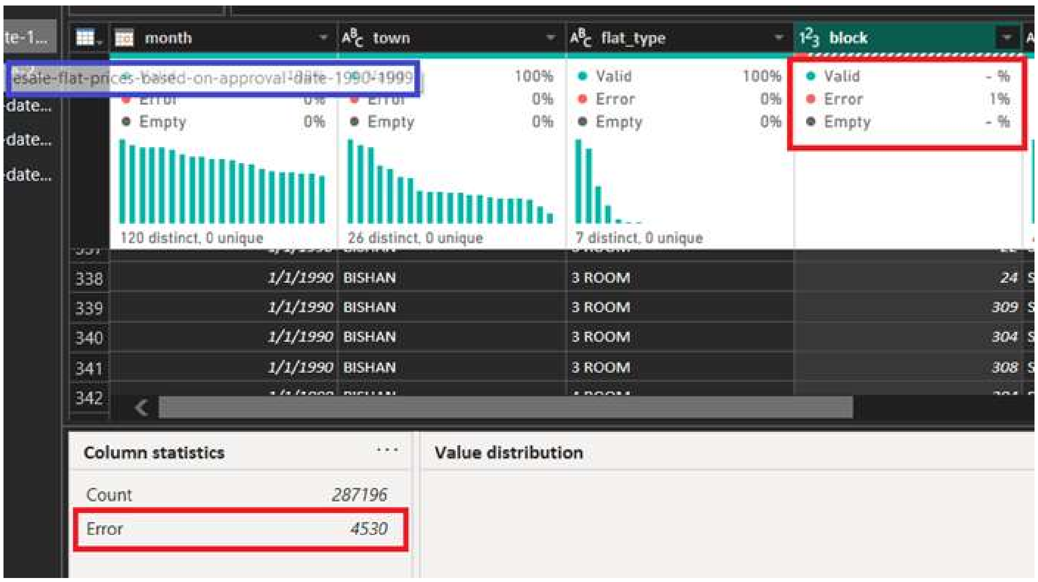
Table: resale-flat-prices-based-on-registration-date-from-jan-2015-to-dec-2016 (37,153 rows)

The resale flat prices Jan 2015 to Dec 2016 table contains 37,153 rows and 11 columns.

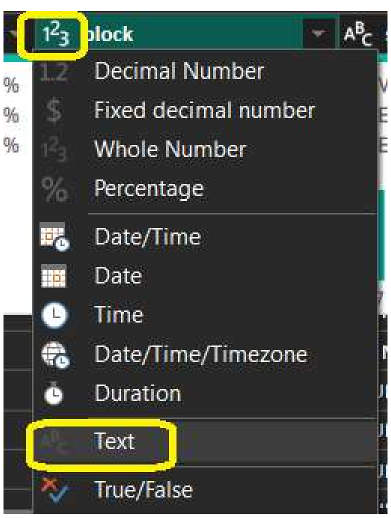
Table: resale-flat-prices-based-on-registration-date-from-jan-2017-onwards (137,207 rows)

The resale flat prices Jan 2017 onwards table contains 137,207 rows and 11 columns.

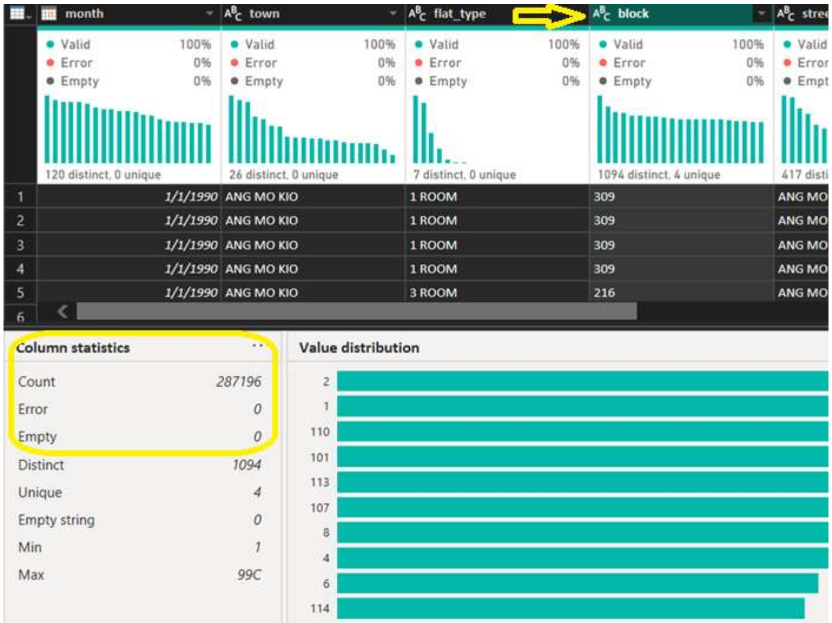
2.2 Transform Data



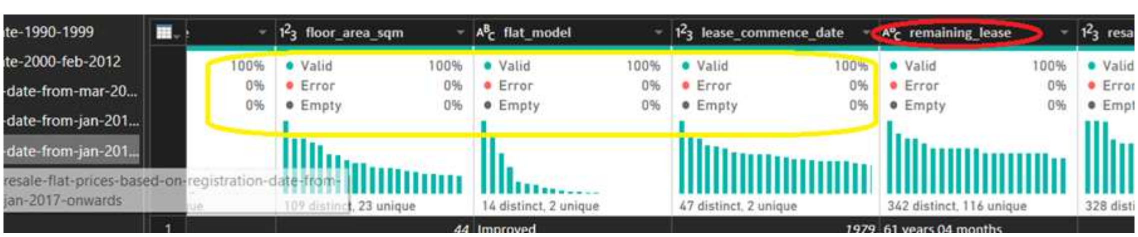
In the block column of the resale flat prices 1990-1999 table, the column profile shows there are 4530 rows with error.



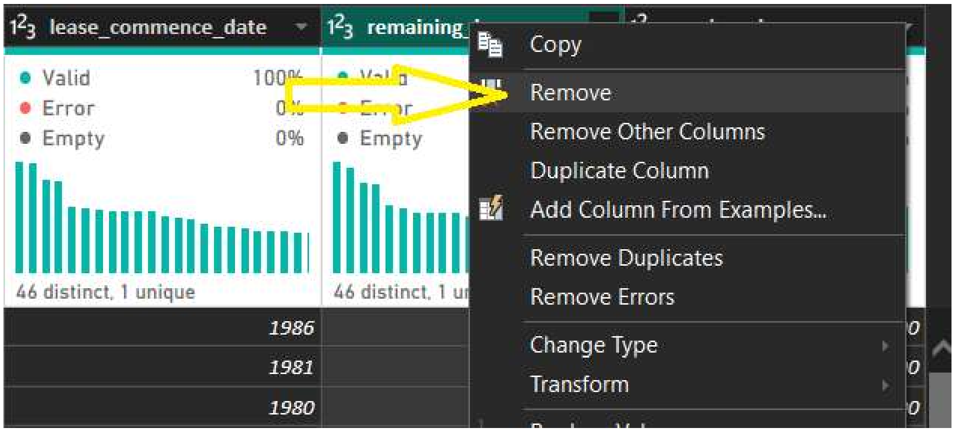
As the column data type is in integer, change it to text.



After changing the datatype to text, the error in the block column was rectified.

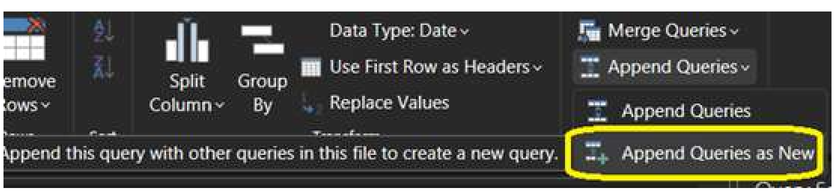


An inspection of the column profile for all five tables shows good data quality. There were two tables 'resale flat prices Jan 2015 to Dec 2016' and 'resale flat prices Jan 2017 onwards' that contains 11 columns. There was an extra 'remaining lease' column. I will remove this column in power query editor and create it after appending the 5 tables into one complete dataset.

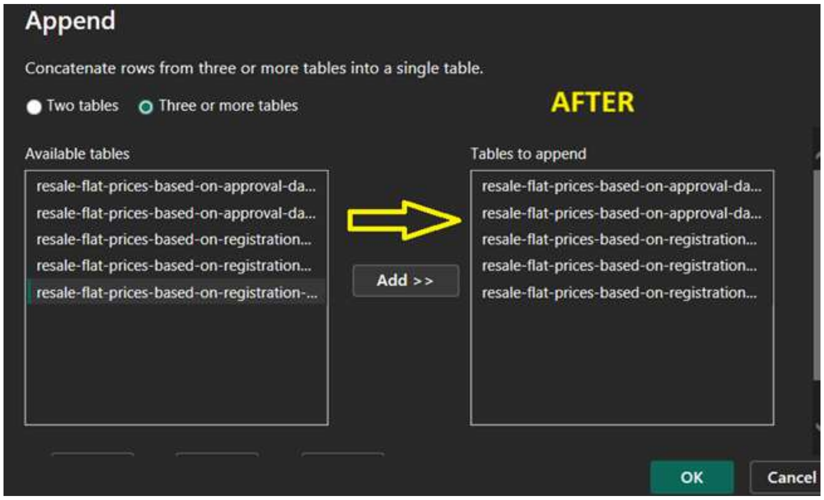


2.3 Append queries

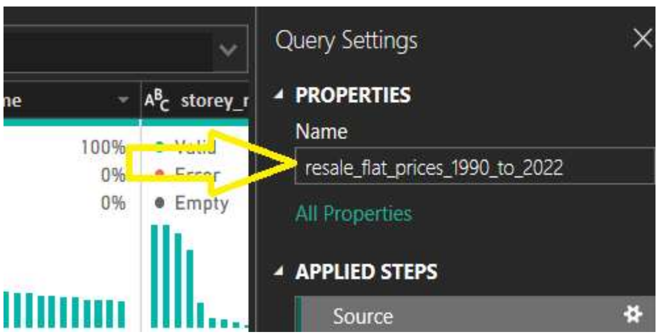
To append the 5 tables in power query editor, go to Append Queries (on the right of the ribbon) and click Append Queries as New. This is recommended as this creates a new combined table called "Append1" and leaves the original tables unchanged.



Select all the tables listed on the left and add them to 'Tables to append'



Rename



The new appended table default name is Append1, I have renamed it to "resale_flat_prices_1990_to_2022"

To check on the total number of rows in this new appended table, go to Transform > Count Rows as demonstrated earlier. This new table resale_flat_prices_1990_to_2022 contains 883,410 rows. Then, close and apply.

2.4 Calculated columns

```
remaining_lease(year) =
// all hdb flats comes with 99-year lease
// subtract flat age from 99 to get the remaining lease as of 2025
99 - (YEAR(TODAY()) - resale_flat_prices_1990_to_2022[lease_commence_date])
```

All HDB flats come with a 99-year lease from the government (gov.sg, 2024). Therefore, to calculate the remaining_lease in 2025 (current year), I have subtracted the lease commence date (in year) from current year (YEAR(TODAY()), and this will become the flat age. To get the remaining lease, I subtract the flat age from 99.

```
floor_area_sqft =
// convert floor area from sqm to sqft
resale_flat_prices_1990_to_2022[floor_area_sqm]*10.764
```

This calculated column floor area sqft was created to convert floor area in sqm to sqft

```
resale_price_per_sqft =
resale_flat_prices_1990_to_2022[resale_price]/resale_flat_prices_1990_to_2022[floor_area_sqft]
```

With a floor area sqft, we can create another calculated column to find the resale price per sqft.

```
flat_age =
//age of flat as of today
YEAR(TODAY()) - resale_flat_prices_1990_to_2022[lease_commence_date]
```

A calculated column to calculate the age of the flat as of today.

```
year_of_transaction =
// Extract year of transaction
YEAR(resale_flat_prices_1990_to_2022[month])
```

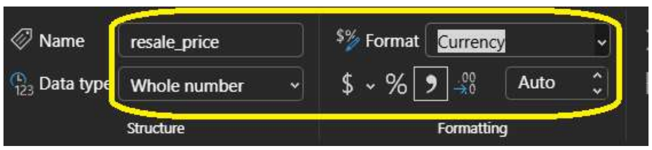
2.5 Measures

```
national_avg_resale_price_per_sqft =
// national average resale price per sqft fixed at storey range, flat type and year of transaction
CALCULATE(
AVERAGE(resale_flat_prices_1990_to_2022[resale_price_per_sqft]), ALLEXCEPT(
// only the columns listed here can affect the calculation hdb_resale_full_raw,
resale_flat_prices_1990_to_2022,
resale_flat_prices_1990_to_2022[storey_range],
resale_flat_prices_1990_to_2022[flat_type],resale_flat_prices_1990_to_2022[year_of_transaction]))
```

Created a measure to calculate the national average resale price per sqft.

```
is_below_national_avg =
// indicate if flat is sold at below national average resale price per sqft
IF(resale_flat_prices_1990_to_2022[resale_price_per_sqft] < [national_avg_resale_price_per_sqft], "Yes", "No")
```

The calculated column ‘is below national avg’



Formatted the resale price column to ‘Currency’


```
resale_price_per_sqft =
resale_flat_prices_1990_to_2022[resale_price]/resale_flat_prices_1990_to_2022[floor_area_sqft]
```

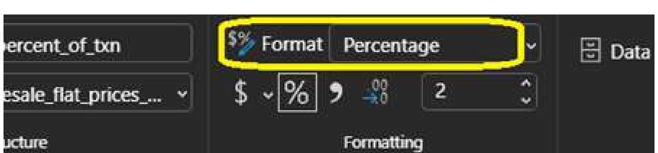
Formatted the resale price per sqft column to ‘Currency’ and 2 decimal places.

```
transaction_count = COUNTROWS(resale_flat_prices_1990_to_2022)
```

Create measure for transaction count, using COUNTROWS function on the entire dataset. This measure is **filter-sensitive**

```
total_txn_count = CALCULATE(COUNTROWS(resale_flat_prices_1990_to_2022),
// calculation is independent of whether the resale price is below national average or not
ALL('resale_flat_prices_1990_to_2022'[is_below_national_avg]))
```

This measure counts every row and does not change even if visuals filter for ‘below national average’ or not.




```
percent_of_txn = [transaction_count]/[total_txn_count]
```

Created another measure percent_of_txn, change the format from “General” to Percentage.

2.6 Summary of Data Preparation

The datasets were relatively clean, there were no missing values and duplicates. Some transformation steps were performed as shown in above pages. One point to highlight was there was a remaining_lease column which was not present in all five tables, this was rectified by creating a calculated column "remaining_lease" for the entire dataset so that insight can be derive from this variable.

Besides data cleaning and transformation, academic research were done to better understand the context of the property scene in Singapore, the Government public housing policies and the economic context during the period 1999 to 2022. The relevant research sources can be found in In-text citation and Reference page.

3. Exploratory Data Analysis and Visualisation

3.1 Univariate Analysis

3.1.1 Descriptive Statistics

resale_price($)		*remaining_lease(year)*		*floor_area_sqft*		*flat_age(year)*	
Mean	308121.8044	Mean	61.79375	Mean	1030.232	Mean	37.20625
Median	287000	Median	60	Median	1001.052	Median	39
Mode	300000	Mode	59	Mode	721.188	Mode	40
Minimum	5000	Minimum	40	Minimum	301.392	Minimum	6
Maximum	1418000	Maximum	93	Maximum	3304.548	Maximum	59
Count	883410	Count	883410	Count	883410	Count	883410

Resale Price Central Tendency

The average (mean) resale price is $308,121.80, slightly higher than the median of $287,000, indicating a mild right-skewed distribution—likely due to million-dollar transactions pulling the mean upward. The mode ($300,000) represents the most frequently occurring price point.

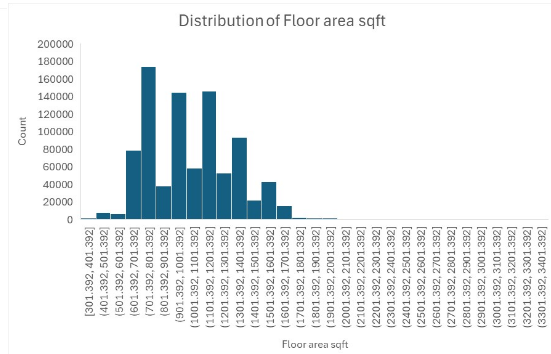
The resale price ranged from $5000 to over $1.4 million, this broad range may not necessary means a diverse market but more akin to economic realities as this dataset spanning over 30 years.

Remaining Lease Central Tendency

The mean of 61.79 years exceeds the median (60 years) and mode (59 years), pointing to a right-skewed distribution. With most leases above 59 years (mode), it signals to potential buyer that HDB resale flats remains viable for long-term ownership

Floor Area Central Tendency

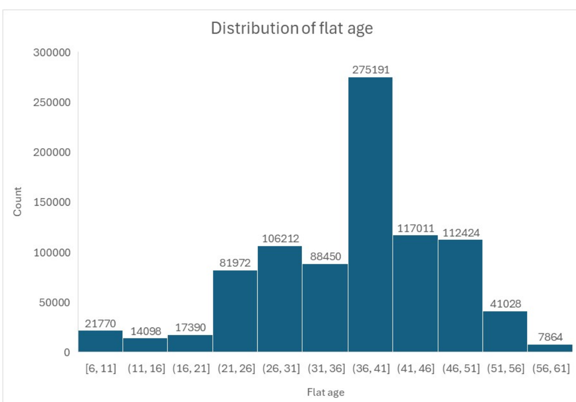
The mean area of 1,030.23 sqft is larger than the median (1,001.05 sqft) and mode (721.19 sqft), confirming right-skewness. The mode indicates smaller, standard 3 room flats are most common, it is also a sign of urban density, where public flats are getting smaller in size and entire precinct of flats were built closer to one another due to land scarcity.

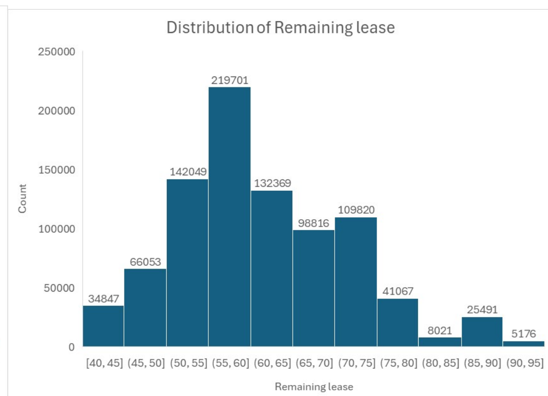


Resale Price: The distribution is **right-skewed**. Most flats cost between $200k and $400k, but there is a long "tail" of expensive flats reaching over $1 million.

Floor Area: This is **multimodal** (has multiple peaks), likely corresponding to standard flat sizes (3-room vs. 4-room vs. 5-room).

Categorical Counts: "4-ROOM" and "3-ROOM" are the most common flat types.

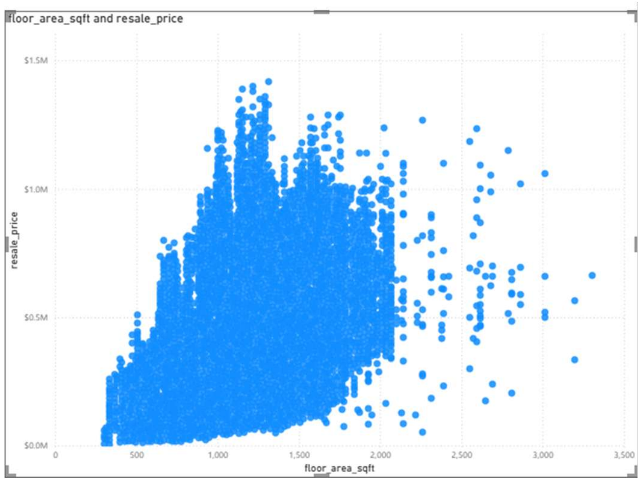


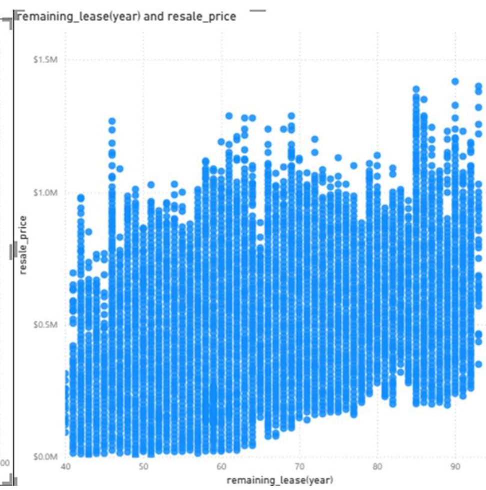


Flat age: The distribution is right-skewed with a prominent peak in the 36-41 year age range (275,191 flats), indicating most flats were built 36-41 years ago. Newer flats (0-25 years) and very old flats (52+ years) are much less common, with the oldest category (56-61 years) having only 7,664 flats.

Remaining Lease years: The distribution shows a bell-shape with a peak in the 50-55 year range (212,701 flats). Most flats have between 45-75 years of remaining lease, representing the bulk of the resale market. The concentration around 50-70 years remaining lease suggests most resale transactions involve flats that are 30-50 years old

3.2 Multivariate Analysis



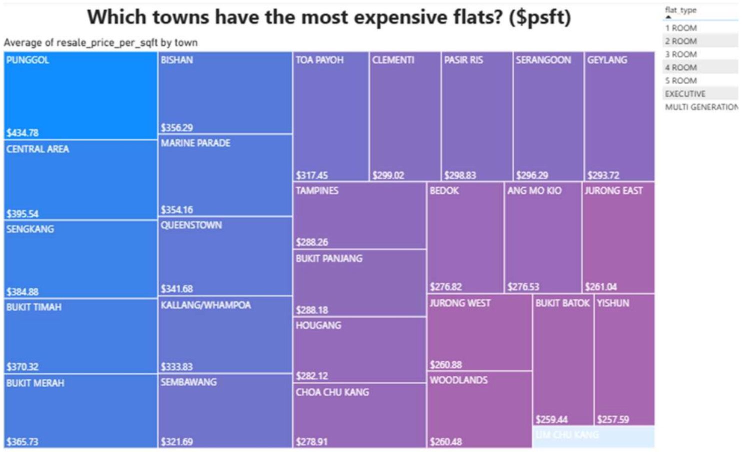


Price vs. Size: There is a strong positive correlation between resale_price and floor_area_sqft. As the flat gets bigger, the price generally goes up.

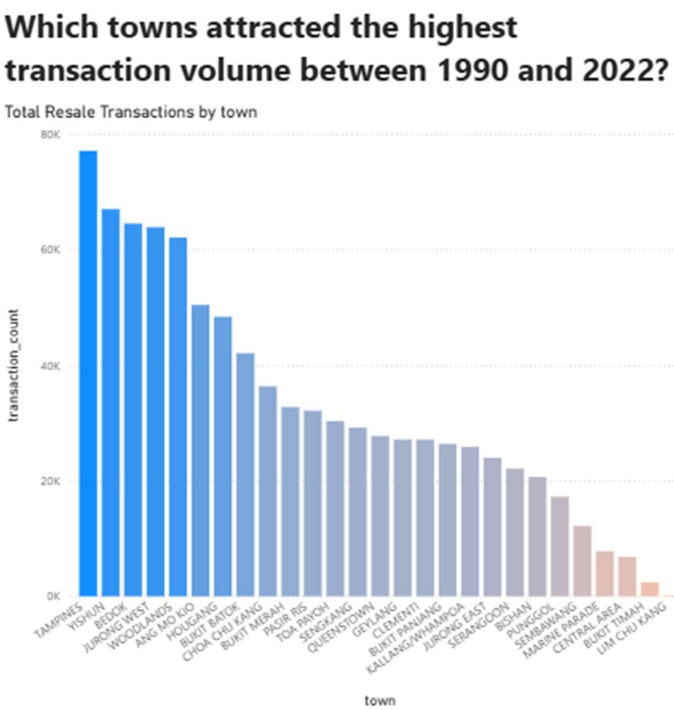
Price vs. Remaining Lease: There is a weaker but noticeable positive relationship. Flats with more years left on the lease generally sell for higher prices.

3.3 Visualisations

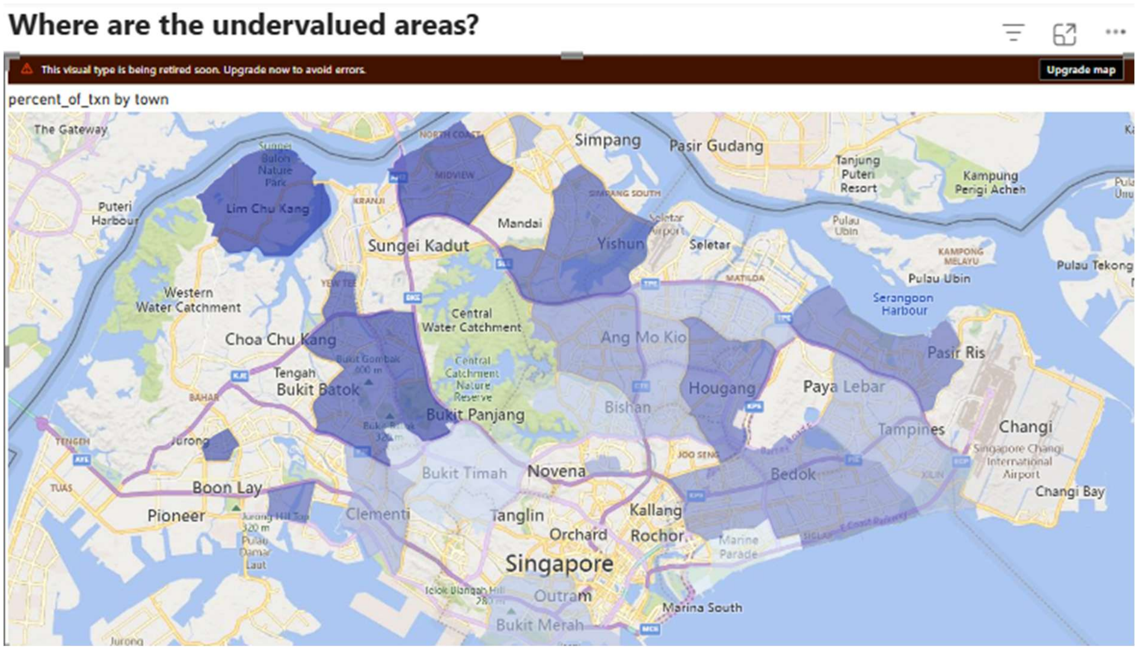
Location based Visuals



This visualisation is a heatmap to explore which towns are the most expensive in terms of average resale price per sqft. The colour gradient allows the audience to quickly spot areas of high and low intensity at a glance without conscious effort. Adding a slicer with flat type will allow the audience to interact with the chart to gain further insights.

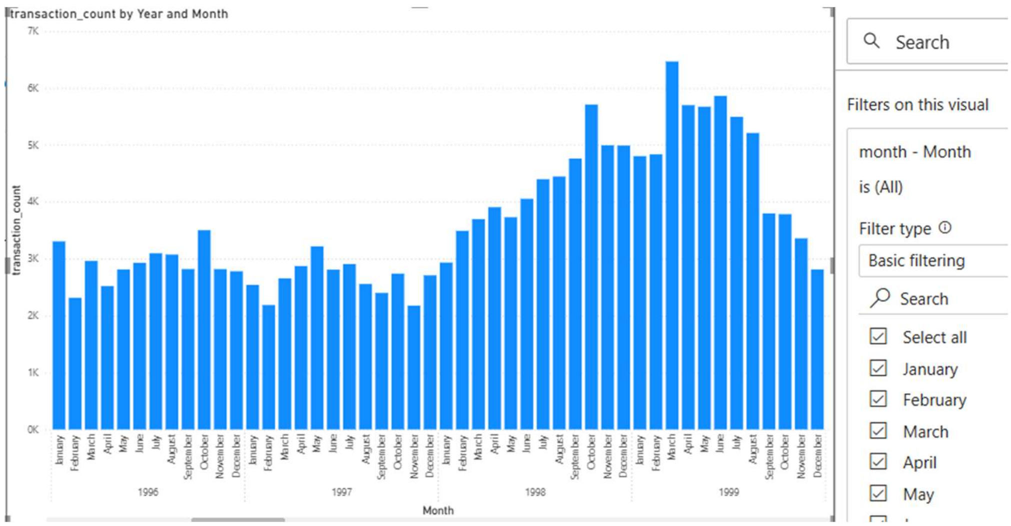


This bar chart shows the total transactions by town. When complement with a flat-type slicer, this chart shows liquidity for which flat-type in which town, i.e. high transaction volume equals easier to buy/sell.

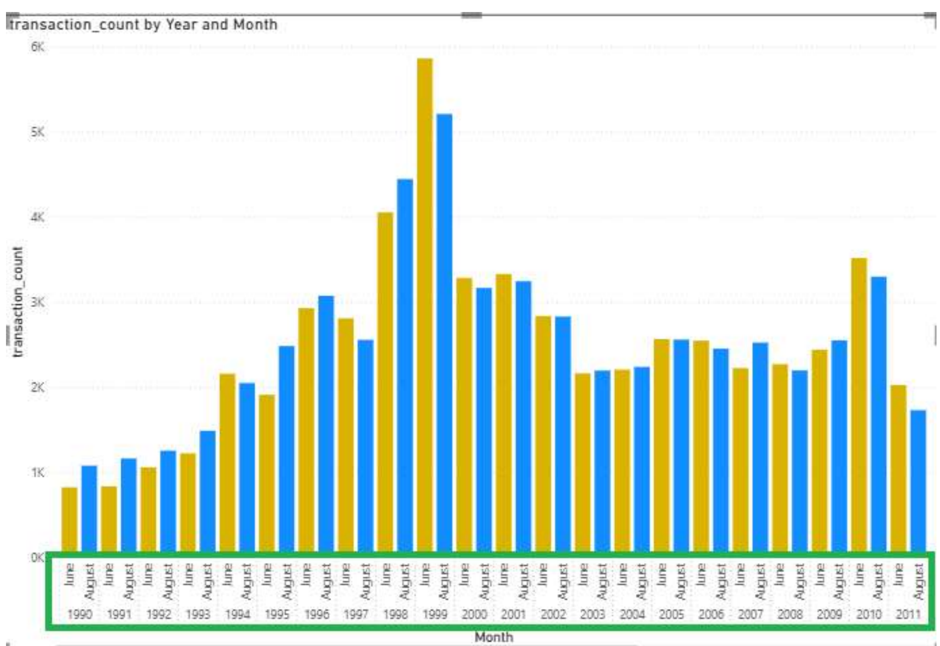


This filled-map visual paints each Singapore town according to **percent_of_txn** of 3-room flats in storey range 4 to 6 that are below the national average. The darker shades indicates where undervalued opportunities appear more frequently and lighter shades the opposite. The viewer can quickly scan the landscape and spot where value resides.

Time Trends

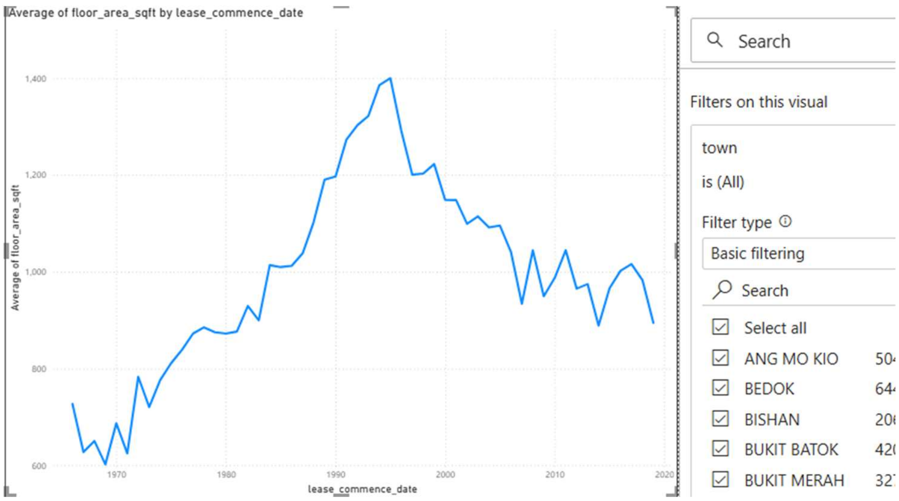


This column bar chart provides insights into whether there is seasonal trend in resale volume. The Asian Financial Crisis started in July 1997 and Singapore weathered the spillover effect through to the second half of 1998 (Chew, 2016). From the transaction volume on this chart, there was a slight dip towards the end of 1997, after that there was a consistent uptrends in transaction volume. As a matter of fact, the highest resale transaction volume could be seen from Oct 1997 to Aug 1998.



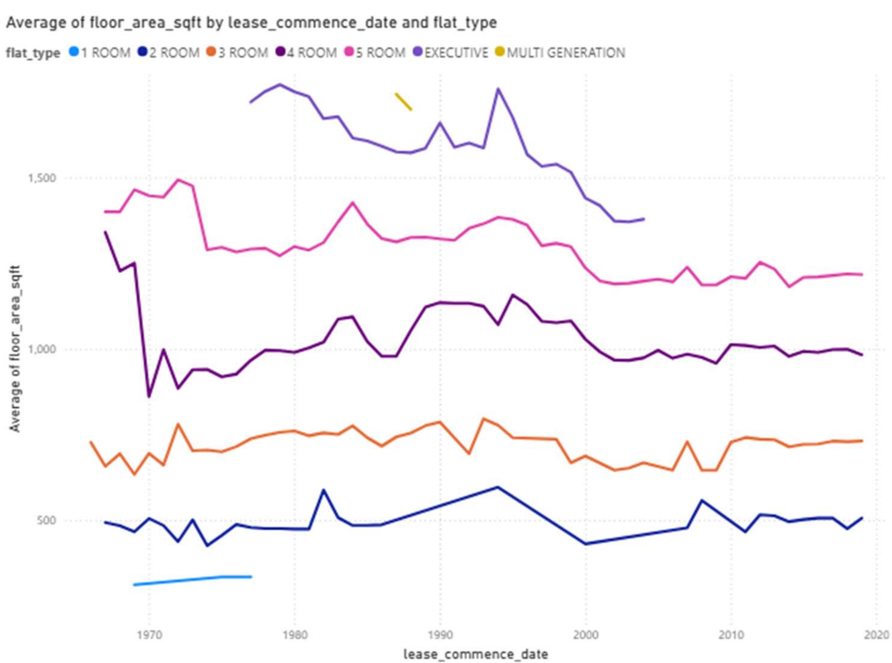
This is the same chart filtered to compare June and August. This can answer question such as "Are there seasonal trends in resale volume? For example, June school holidays or the Hungry Ghost Month?

Flat Features



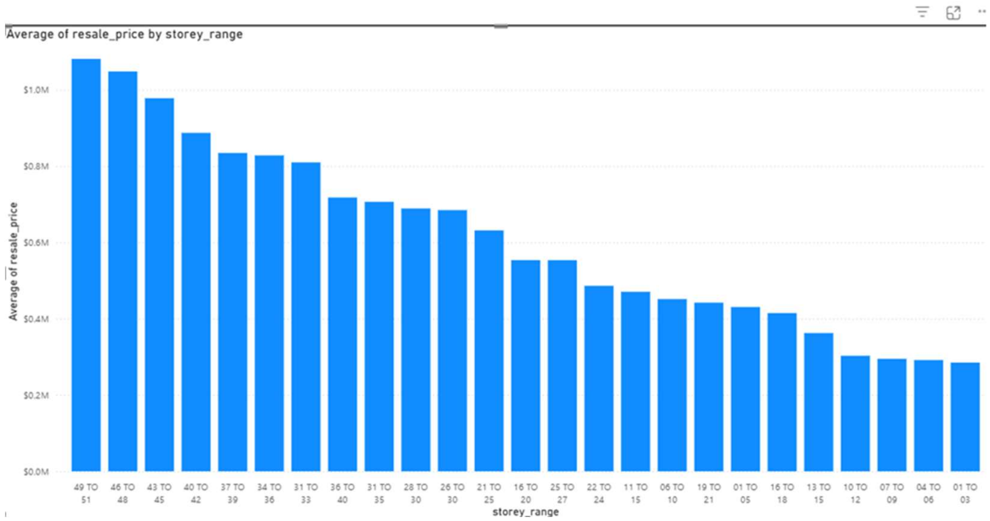
The above line chart shows the trend of average flat size (sqft) for all flat type combine changes over the years. This visual can validate the general consensus that flats are getting smaller and smaller. The average floor area (sqft) peaked in the mid-1990 and has been on a rather sharp downward trend, implying flat size are getting smaller and smaller.

How has the average flat size of each flat type changed over the years?



Drilling down, the average size of each flat type can be visualised in this multiple line chart over the years. Executive flats can be seen as having the most dramatic reduction in average flat size (sqft).

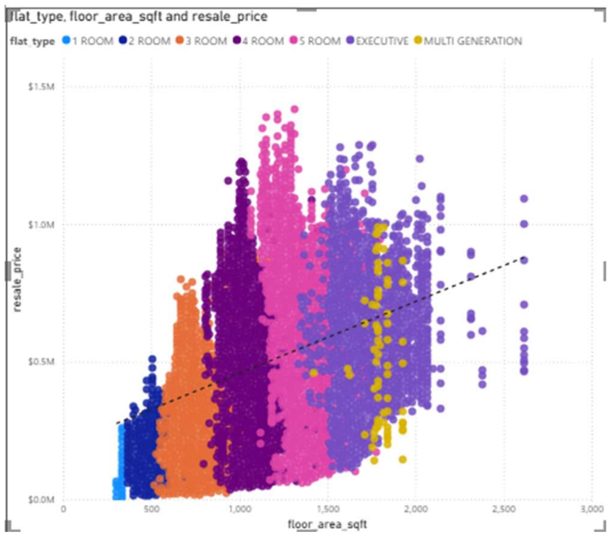
Do Higher Floors Always Command a Premium?



This bar chart shows the **average resale price** across different **storey ranges**, arranged from the highest floors on the left to the lowest floors on the right. The visual confirms a strong **height premium** — the higher the floor, the higher the typical resale price, even surpassing the million-dollar mark.

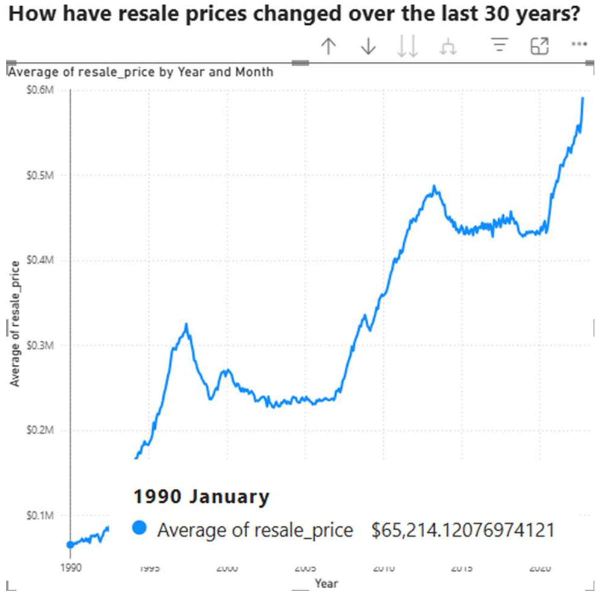
Pricing & Affordability Analysis

How does flat size influence resale price?



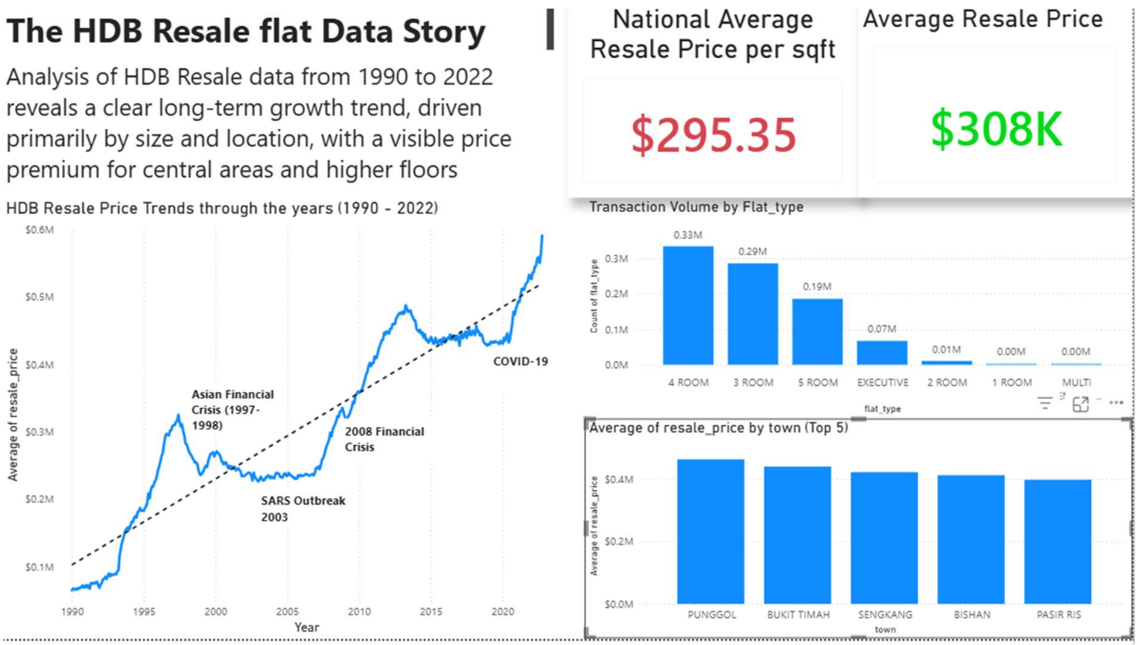
This scatter plot shows the correlation between floor area (sqft) and resale price, with colours distinguishing different **flat types**. Generally, there is a positive correlation, i.e. the bigger the floor area, the higher the resale price. The dashed trend line reinforces the overall positive relationship.

How have resale prices changed over the last 30 years?



This line chart depicts the changes of resale price over the last 30 years. It has been on a dramatic upward trend, save for a few dipping points, which should correspond to some major event affecting the trend. In 1990, the average resale price for all flat type was about $65,000 but the average was close to $600,000 in 2022.

The HDB Resale Flat Data Story: An Overview



Executive Summary: HDB Resale Market Insights (1990–2022)

The HDB Resale flat data from 1990 to 2022 reveals a clear and dramatic long-term growth trend in the average resale price, currently standing at $308K with a National Average Price per Square Foot (PSF) of $295.35. The long-term price trend shows sustained upward momentum, illustrated in the national trend line, reflects steady demand amid economic growth, urbanization, and evolving buyer preferences.

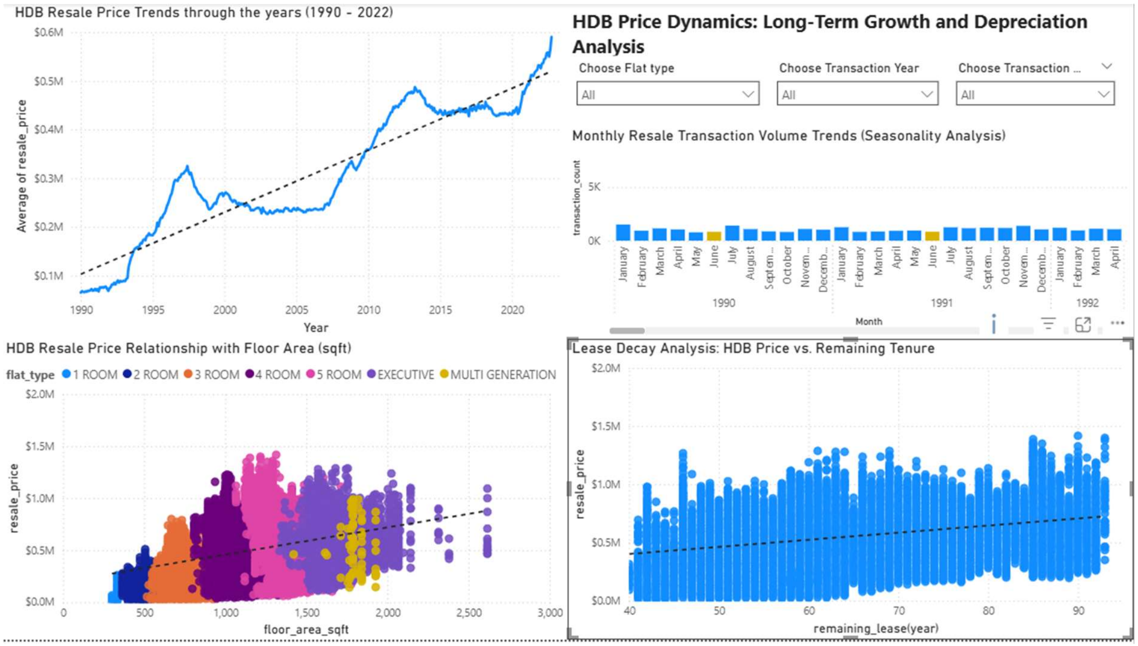
The overall increase is driven primarily by factors of size and location, leading to a visible price premium for central areas and higher floors. In terms of market activity, 4-room and 3-room flats dominate the transaction volume, indicating they are the most liquid segments of the market.

Analysis of the top-performing towns confirms that geographical location remains a prime value driver, with Punggol and Bukit Timah commanding the highest average resale prices. Overall, the data story establishes a strong, positive relationship between flat size and resale price, while the high transaction volume underscores a consistently robust and active market over the analyzed period. This overview serves as a summary of the full data story, which can be further explored through the dedicated Trends, Location, and Features dashboards

4. Dashboard

Dashboard 1: Resale Price Trends

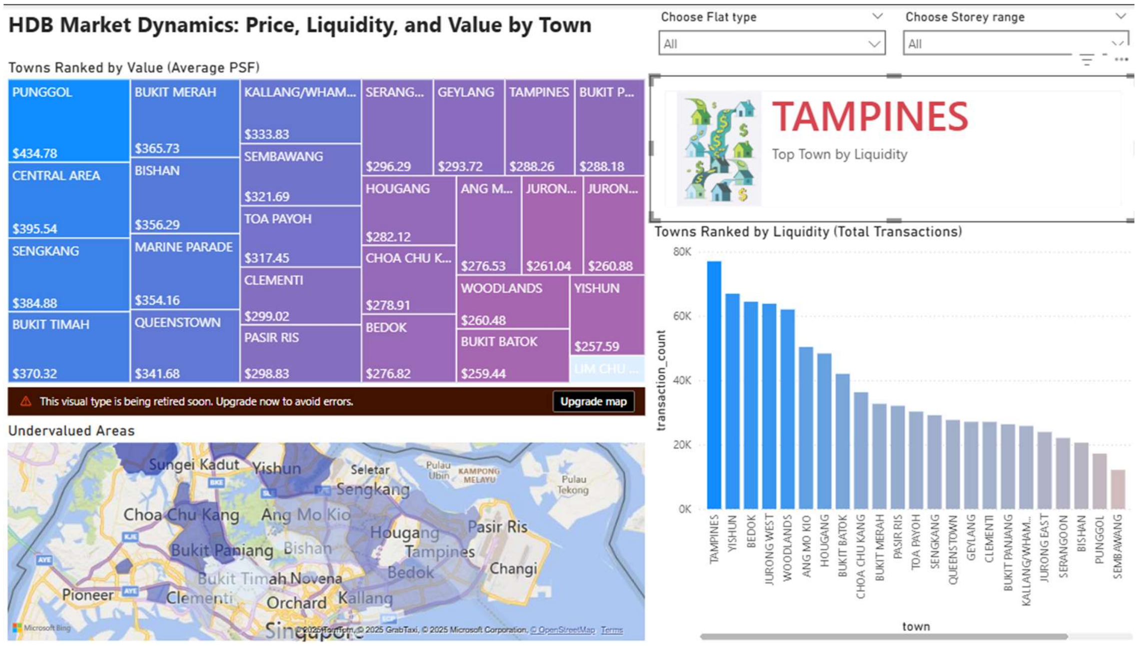
This dashboard focuses on price fluctuations over time, seasonal patterns and lease decay effect on resale price.



The dashboard display the general resale price trends throughout the years. Users can toggle the slicers to discover seasonality patterns, explore relationships between resale price and floor area and see the effect of lease decay.

Dashboard 2: Location-Based Analysis

This dashboard focuses on geographical trends, liquidity, and value.



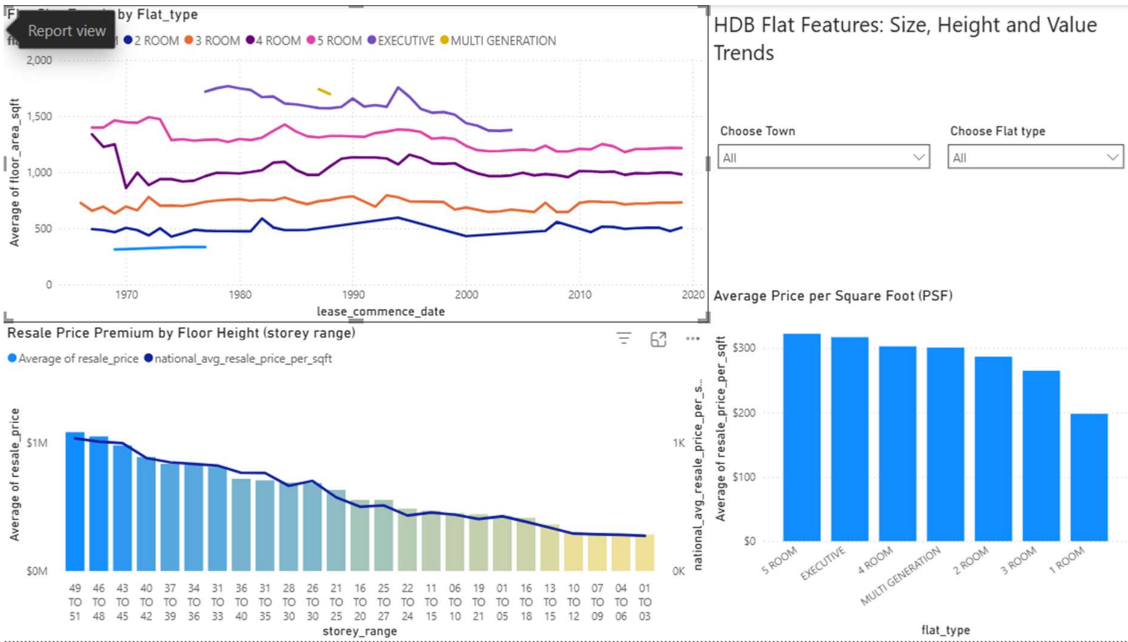
It consist of heatmap showing which towns are the most expensive in terms of average resale price per sqft, a bar chart showing the towns with highest total number of transactions and a filled map showing the areas with percentage of transaction that are below national average.

The slicers of Flat type and Storey range allow users to drill down and see how price and volume change based on specific flat characteristics

This dashboard has all the necessary components for a regular Singaporean property hunter to perform high-quality location analysis.

Dashboard 3: Flat Features

This dashboard focuses on how physical characteristics of the flat (size, type) correlate with price and change over time.



This informative dashboard summarizing key HDB flat features like size, height! It consists of three powerful visuals that work well together: floor area trends, storey-range price premium, and price-per-square-foot (PSF) by flat type. With the addition of flat type and geographical slicers, it allows interactivity and makes the dashboard more insightful.

Reference

gov.sg. (2024, November 21). *A home for everyone: Maintaining the quality and value of your flat*. https://www.gov.sg/explainers/a-home-for-everyone-maintaining-the-quality-and-value-of-your-flat/

Chew, V. (2016, July 30). Asian financial crisis (1997–1998). Singapore Infopedia. National Library Board. https://www.nlb.gov.sg/main/article-detail?cmsuuid=6a94eaac-75ec-41ff-b5ef-38154ccae4e0